

Kinetics and Surface Chemistry Guide

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User prompt: Chemical Kinetics Second order reactions. Determination of order, Fast and slow reaction, steady state approximation, Temperature effect, Concept of Activated Complex / Transition State: Energy of activation, Potential energy surface, Theories of reaction rate: Collision and Transition State theories interms of enzyme catalysis. Surface Chemistry Introduction, Types of adsorption, Adsorption isotherms, BET, Applications of adsorption Give me pdf of all topics cover without leaving any topic

Response: Here is the complete, comprehensive syllabus coverage for **Chemical Kinetics** and **Surface Chemistry**, formatted for your study and structured so it can be exported or printed directly as a PDF (e.g., using your browser's **Print to PDF** feature).

Part I: Chemical Kinetics

1. Second-Order Reactions

A second-order reaction is one where the sum of the exponents in the rate law equals 2.

Case 1: Single Reactant ($2A \rightarrow \text{Products}$) or Identical Initial Concentrations ($A + B \rightarrow \text{Products}$ where $[A]_0 = [B]_0$)

- **Differential Rate Law:**

$$-\frac{d[A]}{dt} = k[A]^2$$

- **Integrated Rate Law:** Separating variables and integrating from $t = 0$ to t :

$$\int_{[A]_0}^{[A]} -\frac{d[A]}{[A]^2} = \int_0^t k dt \implies \frac{1}{[A]} - \frac{1}{[A]_0} = kt$$

$$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$$

- **Half-life ($t_{1/2}$):** At $t = t_{1/2}$, $[A] = \frac{[A]_0}{2}$:

$$\frac{2}{[A]_0} - \frac{1}{[A]_0} = kt_{1/2} \implies t_{1/2} = \frac{1}{k[A]_0}$$

Notice: $t_{1/2}$ is inversely proportional to initial concentration.

- **Units of k :** $\text{L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$ or $\text{M}^{-1}\text{s}^{-1}$.

Case 2: Different Reactants with Unequal Concentrations ($A + B \rightarrow \text{Products}$, $[A]_0 = a$, $[B]_0 = b$, $a \neq b$)

Let x be the amount reacted at time t :

$$\frac{dx}{dt} = k(a - x)(b - x)$$

Using partial fractions and integrating:

$$k = \frac{2.303}{t(a - b)} \log_{10} \frac{b(a - x)}{a(b - x)}$$

2. Determination of Reaction Order

1. **Differential Method (van 't Hoff):** Rate equation: $r = k[A]^n \implies \log r = \log k + n \log[A]$. Plotting $\log r$ vs. $\log[A]$ yields a straight line with slope = n .
2. **Integral / Graphical Method:** Experimental data ($[A]$ vs. t) is tested against integrated rate equations:
 - $[A]$ vs. t is linear $\rightarrow$ Zero order.
 - $\ln[A]$ vs. t is linear $\rightarrow$ First order.
 - $1/[A]$ vs. t is linear $\rightarrow$ Second order.
3. **Half-Life Method ($t_{1/2}$):** General relation: $t_{1/2} \propto \frac{1}{[A]_0^{n-1}}$. Comparing two experiments with initial concentrations $[A]_{0,1}$ and $[A]_{0,2}$:

$$\frac{(t_{1/2})_1}{(t_{1/2})_2} = \left(\frac{[A]_{0,2}}{[A]_{0,1}} \right)^{n-1} \implies n = 1 + \frac{\log[(t_{1/2})_1/(t_{1/2})_2]}{\log[[A]_{0,2}/[A]_{0,1}]}$$

4. **Isolation Method (Ostwald's Method):** For reactions with multiple reactants (e.g., $A + B + C \rightarrow P$), all reactants except one are taken in vast excess. For instance, if $[B]_0, [C]_0 \gg [A]_0$, $[B]$ and $[C]$ remain virtually constant, reducing the rate law to pseudo-order with respect to A . The process is repeated for each reactant.

3. Fast and Slow Reactions & Rate-Determining Step (RDS)

- **Multi-step Reactions:** Complex reactions proceed via a sequence of elementary steps (reaction mechanism).
- **Rate-Determining Step:** The slowest elementary step has the highest activation energy barrier and dictates the overall reaction rate.
- **Methods for Fast Reactions:** Conventional mixing cannot measure reactions faster than milliseconds. Techniques include:
 - **Flow methods:** Continuous and stopped-flow.

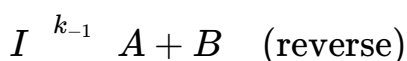
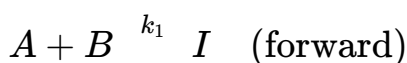
- **Relaxation techniques:** Temperature-jump (T -jump) and pressure-jump (P -jump), where an equilibrium is rapidly perturbed and re-equilibration kinetics are recorded spectrophotometrically.
- **Flash photolysis:** Laser pulses trigger reaction intermediates.

4. Steady-State Approximation (SSA)

For reactive, short-lived intermediates whose concentrations remain low and relatively constant throughout the main course of reaction:

$$\frac{d[\text{Intermediate}]}{dt} \approx 0$$

- **Standard Example:**



Net rate of formation of intermediate I :

$$\frac{d[I]}{dt} = k_1[A][B] - k_{-1}[I] - k_2[I] = 0 \implies [I] = \frac{k_1[A][B]}{k_{-1} + k_2}$$

Overall rate of product formation:

$$\text{Rate} = \frac{d[P]}{dt} = k_2[I] = \frac{k_1 k_2 [A][B]}{k_{-1} + k_2}$$

5. Effect of Temperature on Reaction Rates

Arrhenius Equation

Empirical expression relating rate constant k and temperature T :

$$k = Ae^{-E_a/RT}$$

Where:

- A = Frequency or pre-exponential factor
- E_a = Activation energy ($\text{J} \cdot \text{mol}^{-1}$)
- R = Universal gas constant ($8.314 \text{ J} \cdot \text{mol}^{-1} \text{K}^{-1}$)
- T = Absolute temperature (K)

- **Logarithmic Form:**

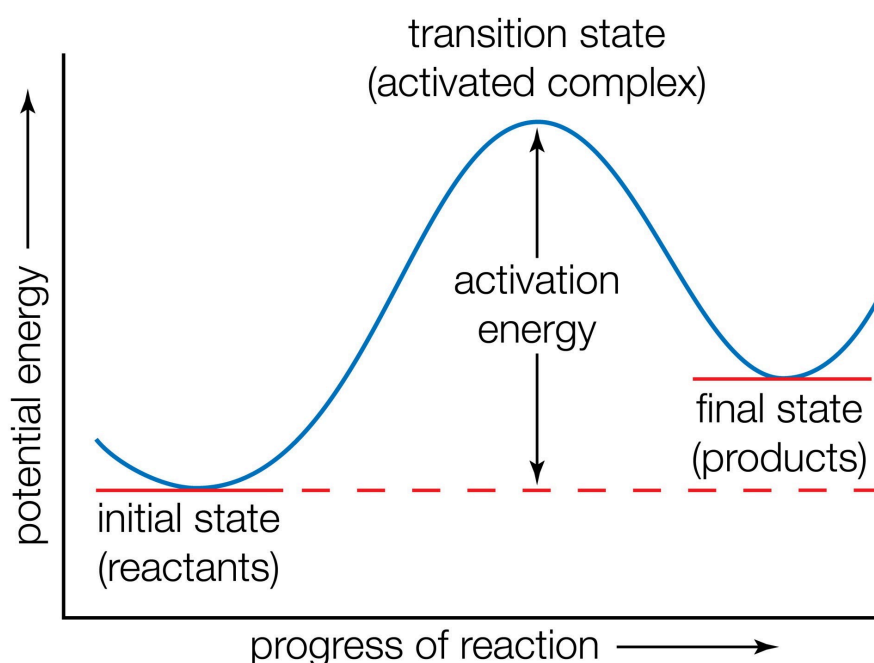
$$\ln k = \ln A - \frac{E_a}{RT} \quad \text{or} \quad \log_{10} k = \log_{10} A - \frac{E_a}{2.303RT}$$

A plot of $\ln k$ versus $1/T$ gives a straight line with slope $= -E_a/R$.

- **Two-Temperature Form:**

$$\ln \left(\frac{k_2}{k_1} \right) = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

6. Concept of Activated Complex & Potential Energy Surfaces



- **Activated Complex / Transition State:** A high-energy, unstable configuration of atoms corresponding to the maximum potential energy along the lowest-energy pathway (reaction coordinate) connecting reactants to products. Old bonds are partially breaking while new bonds are partially forming.
- **Energy of Activation (E_a):** The minimum kinetic energy required by colliding reactant molecules to reach the transition state barrier:

$$E_a = E_{\text{transition state}} - E_{\text{reactants}}$$

- **Potential Energy Surface (PES):** A multi-dimensional mathematical surface mapping potential energy as a function of all nuclear coordinates.
 - Reactants and products reside in local energy minima (valleys).
 - The transition state corresponds to a **saddle point** (maximum along the reaction coordinate, minimum in all orthogonal vibrational directions).

7. Theories of Reaction Rates & Enzyme Catalysis

Collision Theory

- Molecules must collide with sufficient kinetic energy ($\geq$ threshold energy) and correct spatial orientation:

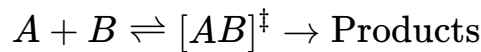
$$k = P \cdot Z_{AB} \cdot e^{-E_a/RT}$$

Where Z_{AB} is collision frequency and P is the steric factor.

- Limitation:** Treats molecules as rigid, hard spheres; fails to account for internal vibrational/rotational degrees of freedom and solvent effects.

Transition State Theory (Eyring Theory)

- Assumes a quasi-equilibrium between reactants and the activated complex:



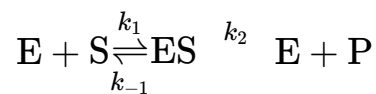
- Eyring-Polanyi Equation:

$$k = \frac{k_B T}{h} K^\ddagger = \frac{k_B T}{h} \exp\left(-\frac{\Delta G^\ddagger}{RT}\right) = \frac{k_B T}{h} \exp\left(\frac{\Delta S^\ddagger}{R}\right) \exp\left(-\frac{\Delta H^\ddagger}{RT}\right)$$

Where k_B is the Boltzmann constant and h is Planck's constant.

Application to Enzyme Catalysis (Michaelis-Menten Kinetics)

Enzymes lower the Gibbs free energy of activation ($\Delta G^\ddagger$) primarily by stabilizing the transition state $[ES]^\ddagger$ through complementary non-covalent interactions (hydrogen bonds, electrostatic charges) within the active site.



Applying Steady-State Approximation to $[ES]$:

$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] = 0$$

Let $[E]_0 = [E] + [ES]$, which yields the **Michaelis-Menten Equation**:

$$v_0 = \frac{V_{\max}[S]}{K_m + [S]}$$

- $V_{\max} = k_2[E]_0 = k_{\text{cat}}[E]_0$
- Michaelis Constant: $K_m = \frac{k_{-1} + k_2}{k_1}$ (substrate concentration at $v_0 = V_{\max}/2$).

Part II: Surface Chemistry

1. Introduction to Adsorption

- **Adsorption:** Accumulation of molecular species at the surface/interface rather than in the bulk of a solid or liquid (a surface phenomenon).
- **Absorption:** Uniform penetration of a substance throughout the bulk of a solid or liquid.
- **Sorption:** Simultaneous occurrence of adsorption and absorption.
- **Desorption:** Process of removing an adsorbed substance from a surface.
- **Thermodynamics:** Adsorption is spontaneously driven ($\Delta G < 0$). Because surface molecules lose translational degrees of freedom upon sticking, entropy decreases ($\Delta S < 0$). Since $\Delta G = \Delta H - T\Delta S$, adsorption is **always exothermic** ($\Delta H < 0$).

2. Types of Adsorption: Physisorption vs. Chemisorption

Property	Physisorption (Physical Adsorption)	Chemisorption (Chemical Adsorption)
Forces	Weak van der Waals forces	Strong chemical (covalent/ionic) bonds
Enthalpy of Adsorption	Low (20 – 40 kJ/mol)	High (80 – 400 kJ/mol)
Reversibility	Reversible	Usually irreversible
Specificity	Non-specific	Highly specific
Layer Formation	Multilayer formation at high pressure	Monolayer only
Temperature Effect	Decreases monotonically as T rises	Increases first (requires E_a), then decreases
Activation Energy	Almost zero	Substantial activation energy required

3. Adsorption Isotherms

An adsorption isotherm expresses the variation in the amount of gas adsorbed by a unit mass of adsorbent with pressure at constant temperature.

Freundlich Adsorption Isotherm

Empirical relationship for heterogeneous surfaces:

$$\frac{x}{m} = kP^{1/n} \quad (n > 1)$$

Taking the logarithm:

$$\log \left(\frac{x}{m} \right) = \log k + \frac{1}{n} \log P$$

- At low pressure: $1/n = 1 \implies x/m \propto P$.
- At high pressure: $1/n = 0 \implies x/m = \text{constant}$ (saturation).
- **Limitation:** Fails at high pressure; purely empirical without theoretical basis.

Langmuir Adsorption Isotherm

Based on kinetic equilibrium between condensation (adsorption) and evaporation (desorption) on a surface.

- **Key Postulates:**
 1. Adsorption is limited to a monolayer.
 2. All surface adsorption sites are energetically equivalent (homogeneous).
 3. No lateral interactions exist between adjacent adsorbed molecules.
- **Derivation:** Let θ be the fraction of covered surface, and $(1 - \theta)$ the fraction of bare surface.
 - Rate of adsorption: $r_a = k_a P(1 - \theta)$
 - Rate of desorption: $r_d = k_d \theta$ At dynamic equilibrium ($r_a = r_d$):

$$k_a P(1 - \theta) = k_d \theta \implies \theta = \frac{bP}{1 + bP} \quad \left(\text{where } b = \frac{k_a}{k_d} \right)$$

Since $x/m \propto \theta$:

$$\frac{x}{m} = \frac{aP}{1 + bP}$$

- **Linear Form:**

$$\frac{P}{x/m} = \frac{1}{a} + \frac{b}{a}P$$

4. Brunauer-Emmett-Teller (BET) Theory

BET theory extends the Langmuir mechanism to **multilayer adsorption** for physical adsorption.

- **Key Assumptions:**

1. Gas molecules adsorb on solid surfaces in successive layers indefinitely.
2. The Langmuir equation applies to each individual layer.
3. First layer has a characteristic heat of adsorption (ΔH_{ads}).
4. Second and higher layers behave like a condensed liquid; their heats of adsorption equal the heat of liquefaction/condensation (ΔH_L).

• **BET Equation:**

$$\frac{P}{V(P_0 - P)} = \frac{1}{V_m C} + \frac{C - 1}{V_m C} \left(\frac{P}{P_0} \right)$$

Where:

- P = Equilibrium pressure; P_0 = Saturation vapor pressure of the gas at adsorption temperature.
 - V = Total volume of gas adsorbed at STP.
 - V_m = Monolayer volume capacity at STP.
 - C = Dimensionless BET parameter $\approx \exp \left(\frac{\Delta H_{\text{ads}} - \Delta H_L}{RT} \right)$.
- **Determination of Specific Surface Area:** Plotting $\frac{P}{V(P_0 - P)}$ vs. $\frac{P}{P_0}$ yields slope $s = \frac{C-1}{V_m C}$ and intercept $i = \frac{1}{V_m C}$.

$$V_m = \frac{1}{s + i}$$

Total surface area (S_{total}) and specific surface area (S_{BET}):

$$S_{\text{BET}} = \frac{V_m \cdot N_A \cdot \sigma}{V_{\text{molar}} \cdot m}$$

(N_A = Avogadro's number, σ = molecular cross-sectional area of the adsorbate, $V_{\text{molar}} = 22,400 \text{ cm}^3/\text{mol}$).

5. Industrial & Laboratory Applications of Adsorption

1. **Production of Ultra-High Vacuum:** Residual traces of air inside evacuated vessels are captured using activated charcoal cooled in liquid air.
2. **Gas Masks:** Granular activated carbon or metal-oxide sorbents selectively adsorb hazardous gases (Cl_2 , phosgene, carbon monoxide) while letting oxygen pass through.
3. **Humidity Control and Desiccation:** Silica gel and activated alumina pellets remove moisture from enclosed electronic hardware, pharmaceuticals, and packaged goods.
4. **Decolorization and Refining:** Animal charcoal or activated clay removes colored organic impurities from sugar syrups, vegetable oils, and petroleum fractions.

5. **Heterogeneous Catalysis:** Gaseous reactants adsorb onto solid metal catalysts (Fe in the Haber-Bosch ammonia synthesis, Ni in hydrogenation of oils, V_2O_5 in the Contact process), reducing activation energy.
 6. **Chromatographic Separations:** Column, thin-layer, and gas-solid chromatography separate complex chemical mixtures based on differences in adsorption affinities.
 7. **Ion-Exchange Resins:** Polymeric synthetic resins exchange mobile functional ions to deionize water and soften industrial boiler feeds.
 8. **Froth Flotation Process:** Low-grade sulfide ores are selectively wetted with pine oil and collected in the foam layer, separating them from gangue minerals.
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Response: Comprehensive Study Guide: Chemical Kinetics and Surface Chemistry

Part 1: Chemical Kinetics

1. Second-Order Reactions

A reaction is second-order when the rate is proportional to either the square of the concentration of a single reactant or the product of the concentrations of two different reactants.

Case A: Single Reactant ($2A \rightarrow \text{Products}$)

The differential rate law is:

$$-\frac{d[A]}{dt} = k[A]^2$$

Separating variables and integrating from $t = 0$ ($[A] = [A]_0$) to time t ($[A] = [A]_t$):

$$\int_{[A]_0}^{[A]_t} -\frac{d[A]}{[A]^2} = k \int_0^t dt \implies \left[\frac{1}{[A]} \right]_{[A]_0}^{[A]_t} = kt$$
$$\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$$

- **Linear Plot:** A plot of $\frac{1}{[A]_t}$ versus t gives a straight line with slope = k and y-intercept = $\frac{1}{[A]_0}$.
- **Half-life ($t_{1/2}$):** Setting $[A]_t = \frac{[A]_0}{2}$:

$$\frac{2}{[A]_0} - \frac{1}{[A]_0} = kt_{1/2} \implies t_{1/2} = \frac{1}{k[A]_0}$$

Unlike first-order reactions, the half-life is inversely proportional to the initial concentration.

- **Rate Constant Units:** $L \cdot mol^{-1} \cdot s^{-1}$ (or $M^{-1}s^{-1}$).

Case B: Two Different Reactants ($A + B \rightarrow$ Products)

With rate law $-\frac{d[A]}{dt} = k[A][B]$ and initial concentrations $[A]_0 \neq [B]_0$:

$$\frac{1}{[A]_0 - [B]_0} \ln \left(\frac{[B]_0[A]_t}{[A]_0[B]_t} \right) = kt$$

2. Determination of Reaction Order

Method	Principle	Mathematical Formulation / Procedure
Differential Method (van 't Hoff)	Compares instantaneous rates at varying concentrations.	$\text{Rate} = k[A]^n \implies \log(\text{Rate}) = \log k + n \log[A]$. The slope of $\log(\text{Rate})$ vs. $\log[A]$ yields order n .
Integral (Graphical) Method	Fit experimental concentration-time data into integrated rate equations.	Data is plotted for zero-order ($[A]$ vs t), first-order ($\ln[A]$ vs t), and second-order ($\frac{1}{[A]}$ vs t). The plot displaying linearity dictates the order.
Half-life Method	Evaluates dependence of half-life on initial reactant concentration.	$t_{1/2} \propto \frac{1}{[A]_0^{n-1}}$. Using two runs: $n = 1 + \frac{\log(t_{1/2,(1)}/t_{1/2,(2)})}{\log([A]_{0,(2)}/[A]_{0,(1)})}$.
Ostwald's Isolation Method	Isolates the dependence on a single reactant by flooding the system with all other reactants.	If reaction is $A + B + C \rightarrow P$, run with large excess of B and C ($[B], [C] \approx \text{const}$). The apparent rate = $k'[A]^\alpha$, yielding partial order α .

3. Fast and Slow Reactions

- **Slow/Moderate Reactions:** Half-times range from seconds to days (e.g., ester hydrolysis, sugar inversion). Monitored using standard spectrophotometry, titrimetry, conductometry, or polarimetry.

- **Fast Reactions:** Reactions with half-lives ranging from milliseconds down to femtoseconds (10^{-3} s to 10^{-15} s), such as acid-base neutralizations ($H^+ + OH^- \rightarrow H_2O$) and radical recombinations. Diffusion often limits these processes.

Experimental Techniques for Fast Reactions

1. Flow Methods:

- *Continuous Flow:* Reactants are continuously driven through a mixing chamber; distance along the observation tube corresponds to time.
- *Stopped-Flow:* Reactants are rapidly forced into a mixing chamber and suddenly stopped in an observation cell. Dead time is roughly 1 ms.

2. Relaxation Methods: A system at equilibrium is perturbed extremely rapidly by an external thermodynamic variable, and the relaxation to a new equilibrium state is tracked:

- *Temperature Jump (T-jump):* Rapid discharge of a capacitor raises temperature by 5–10 K within nanoseconds.
- *Pressure Jump (P-jump):* Rapid burst of a diaphragm changes pressure in sub-milliseconds.

3. Flash Photolysis & Pump-Probe Spectroscopy: A laser pulse (pump) initiates reaction by photochemical dissociation; a second delayed pulse (probe) monitors transient intermediate absorption down to femtosecond resolution.

4. Steady-State Approximation (SSA)

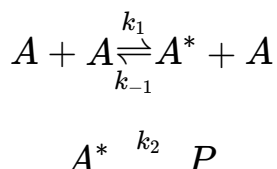
In complex multi-step reaction mechanisms involving reactive, short-lived intermediates, the intermediate concentration remains small and nearly constant throughout the bulk of the reaction.

- **Mathematical Condition:** For a reactive intermediate I :

$$\frac{d[I]}{dt} \approx 0$$

$$\text{Rate of Formation of } I = \text{Rate of Consumption of } I$$

Example: Lindemann-Hinshelwood Mechanism for Unimolecular Reactions



Applying SSA to energized molecule A^* :

$$\frac{d[A^*]}{dt} = k_1[A]^2 - k_{-1}[A^*][A] - k_2[A^*] = 0$$

$$[A^*] = \frac{k_1[A]^2}{k_{-1}[A] + k_2}$$

The overall rate of product formation is:

$$\text{Rate} = k_2[A^*] = \frac{k_1 k_2 [A]^2}{k_{-1}[A] + k_2}$$

- **High Pressure Limit** ($k_{-1}[A] \gg k_2$): $\text{Rate} \approx \frac{k_1 k_2}{k_{-1}} [A]$ (First-order kinetics).
- **Low Pressure Limit** ($k_2 \gg k_{-1}[A]$): $\text{Rate} \approx k_1 [A]^2$ (Second-order kinetics).

5. Temperature Effect and Arrhenius Equation

Reaction rates increase sharply with increasing temperature (typically doubling or tripling per 10°C rise near room temperature).

Arrhenius Equation

$$k = A \exp \left(-\frac{E_a}{RT} \right)$$

$$\ln k = \ln A - \frac{E_a}{RT}$$

- k : Rate constant
- A : Pre-exponential (frequency) factor, representing collision frequency and orientation probability
- E_a : Activation energy ($\text{J} \cdot \text{mol}^{-1}$)
- R : Universal gas constant ($8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$)
- T : Absolute temperature (K)

Differentiating with respect to temperature gives:

$$\frac{d \ln k}{dT} = \frac{E_a}{RT^2}$$

Between two temperatures T_1 and T_2 :

$$\ln \left(\frac{k_2}{k_1} \right) = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

6. Activated Complex / Transition State

Potential Energy (V)

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[Activated Complex / TS (‡)]



- **Energy of Activation (E_a):** The minimum kinetic energy colliding reactant molecules must possess along the line of centers to overcome inter-nuclear repulsion and distort chemical bonds into the transition state.
- **Activated Complex (Transition State, $\ddagger$):** An ultra-short-lived ($\sim 10^{-13}$ s) molecular entity located at the maximum potential energy along the lowest-energy reaction path. It contains partially broken and partially formed bonds and cannot be isolated.
- **Potential Energy Surface (PES):** A mathematical/graphical representation of molecular potential energy as a function of the geometric coordinates of the atomic nuclei.
 - Reactants and products reside in local energy minima.
 - The **Minimum Energy Path (MEP)** or intrinsic reaction coordinate connects reactants to products across a col or **saddle point**.
 - At the saddle point, the energy is a maximum along the reaction coordinate, but a minimum along all perpendicular coordinates.

7. Theories of Reaction Rate

A. Collision Theory

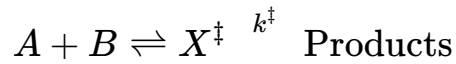
Applies the kinetic theory of hard-sphere gases to bimolecular reactions:

$$\text{Rate} = Z_{AB} \cdot \rho \cdot \exp\left(-\frac{E_a}{RT}\right)$$

- Z_{AB} : Collision frequency between molecules A and B .
- ρ (or P): Steric factor (probability factor, often $\ll 1$) accounting for the requirement that molecules collide with the correct spatial orientation.
- $\exp(-E_a/RT)$: Boltzmann fraction of collisions possessing energy $\geq E_a$.
- *Limitation*: Treats molecules as rigid, structureless spheres and fails to predict ρ theoretically for polyatomic systems without empirical fitting.

B. Transition State Theory (Eyring-Polanyi Theory)

Assumes a quasi-equilibrium between reactants and the activated complex ($X^\ddagger$), which decomposes into products at a frequency proportional to fundamental vibration:



The rate constant is given by the Eyring equation:

$$k = \frac{k_B T}{h} K^\ddagger$$

From chemical thermodynamics ($\Delta G^\ddagger = -RT \ln K^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger$):

$$k = \frac{k_B T}{h} \exp\left(\frac{\Delta S^\ddagger}{R}\right) \exp\left(-\frac{\Delta H^\ddagger}{RT}\right)$$

- k_B : Boltzmann constant ($1.38 \times 10^{-23} \text{ J} \cdot \text{K}^{-1}$)
- h : Planck constant ($6.626 \times 10^{-34} \text{ J} \cdot \text{s}$)
- $\Delta G^\ddagger$, $\Delta H^\ddagger$, $\Delta S^\ddagger$: Gibbs free energy, enthalpy, and entropy of activation.

C. Transition State Theory in Enzyme Catalysis

Enzymes accelerate biochemical reactions by factors of 10^6 to 10^{17} .

1. Michaelis-Menten Mechanism:



Applying SSA to $[ES]$:

$$v = -\frac{d[S]}{dt} = \frac{V_{max}[S]}{K_m + [S]} = \frac{k_{cat}[E]_0[S]}{K_m + [S]}$$

Where $K_m = \frac{k_{-1} + k_{cat}}{k_1}$ (Michaelis constant) and $V_{max} = k_{cat}[E]_0$.

2. TST Formulation of Enzymatic Power: According to transition state theory, catalytic rate constants relate directly to activation free energy:

$$k_{uncat} = \frac{k_B T}{h} \exp\left(-\frac{\Delta G_{uncat}^\ddagger}{RT}\right)$$

$$k_{cat} = \frac{k_B T}{h} \exp\left(-\frac{\Delta G_{cat}^\ddagger}{RT}\right)$$

$$\frac{k_{cat}}{k_{uncat}} = \exp\left(\frac{\Delta\Delta G^\ddagger}{RT}\right) \quad \text{where } \Delta\Delta G^\ddagger = \Delta G_{uncat}^\ddagger - \Delta G_{cat}^\ddagger$$

- **Transition State Stabilization:** As proposed by Linus Pauling, an enzyme active site is structurally and electrostatically complementary to the **transition state** ($S^\ddagger$), rather

than the ground-state substrate (S).

- By binding the transition state with vastly higher affinity than the substrate ($K_T \ll K_S$), the enzyme lowers $\Delta G^\ddagger$, driving enzymatic rate enhancement.

Part 2: Surface Chemistry

1. Introduction to Adsorption

Adsorption is the spontaneous phenomenon of higher concentration/accumulation of molecular species (adsorbate) at the surface or interface of a solid or liquid (adsorbent) relative to the bulk phase.

- **Cause:** Unsatisfied, unbalanced residual molecular forces or free valencies at the boundary surface pull surface molecules inward, giving rise to surface tension and surface free energy. Adsorption reduces this surface energy ($\Delta G_{ads} < 0$).
- **Thermodynamics:**

$$\Delta G_{ads} = \Delta H_{ads} - T \Delta S_{ads}$$

Since adsorption restricts adsorbate degrees of freedom, $\Delta S_{ads} < 0$. For ΔG_{ads} to remain negative, ΔH_{ads} **must always be negative (exothermic)**.

2. Types of Adsorption: Physisorption vs. Chemisorption

Characteristic	Physical Adsorption (Physisorption)	Chemical Adsorption (Chemisorption)
Origin of Forces	Weak van der Waals forces (dipole-dipole, dispersion).	Strong chemical bonds (covalent, ionic, coordinate).
Enthalpy of Adsorption (ΔH_{ads})	Low: 20 to 40 kJ · mol ⁻¹ .	High: 80 to 400 kJ · mol ⁻¹ .
Specificity	Non-specific; occurs on any solid surface.	Highly specific; requires chemical affinity.
Reversibility	Readily reversible by decreasing pressure or increasing temperature.	Irreversible; desorbed species often chemically altered.
Layer Thickness	Multimolecular layers form under high pressure.	Unimolecular (monolayer) strictly.
Activation Energy (E_a)	Negligible activation energy required.	Often requires substantial activation energy.
Effect of Temperature	Decreases continuously as temperature rises.	First increases (overcoming E_a), then decreases.

3. Adsorption Isotherms

An adsorption isotherm models the mathematical relationship between the amount of gas adsorbed (x/m) and the equilibrium gas pressure (P) at constant temperature.

A. Freundlich Adsorption Isotherm

An empirical equation modeling adsorption on heterogeneous surfaces:

$$\frac{x}{m} = kP^{1/n} \quad (n > 1)$$

- x : Mass of adsorbate; m : Mass of adsorbent; P : Equilibrium pressure.
- k, n : Empirical constants depending on the nature of the gas, adsorbent, and temperature.

Taking the logarithm of both sides gives a linear form:

$$\log \left(\frac{x}{m} \right) = \log k + \frac{1}{n} \log P$$

- **Plot:** $\log(x/m)$ vs. $\log P$ yields a straight line with slope = $\frac{1}{n}$ (ranging between 0 and 1) and intercept = $\log k$.
- **Limitations:** Fails at elevated pressures where adsorption approaches saturation.

B. Langmuir Adsorption Isotherm

Derived from kinetic considerations of adsorption and desorption.

Fundamental Postulates:

1. The adsorbent surface consists of fixed, uniform elementary sites of equal energy.
2. Each site can hold at most one adsorbate molecule (monomolecular coverage).
3. No lateral interactions exist between adjacent adsorbed molecules.
4. Dynamic equilibrium exists between adsorption and desorption.

Derivation:

Let surface coverage fraction be θ ; vacant site fraction is $(1 - \theta)$.

- Rate of Adsorption = $k_a P(1 - \theta)$
- Rate of Desorption = $k_d \theta$

At dynamic equilibrium:

$$k_a P(1 - \theta) = k_d \theta$$

$$k_a P = \theta(k_d + k_a P) \implies \theta = \frac{k_a P}{k_d + k_a P} = \frac{bP}{1 + bP}$$

Where $b = \frac{k_a}{k_d}$ is the adsorption equilibrium constant.

Since $\frac{x}{m} \propto \theta \implies \frac{x}{m} = \frac{aP}{1+bP}$:

$$\frac{P}{x/m} = \frac{1}{a} + \frac{b}{a}P$$

- **Plot:** $\frac{P}{x/m}$ versus P gives a straight line with slope $= \frac{b}{a}$ and intercept $= \frac{1}{a}$.
- **Low Pressure Limit** ($bP \ll 1$): $\frac{x}{m} \approx aP$ (First-order dependence on pressure).
- **High Pressure Limit** ($bP \gg 1$): $\frac{x}{m} \approx \frac{a}{b} = \text{constant}$ (Zero-order kinetics, complete monolayer saturation).

4. BET (Brunauer-Emmett-Teller) Theory

The BET theory extends Langmuir's kinetic model to explain **multilayer physical adsorption**, which occurs when gas molecules condense onto previously adsorbed layers.

Key Assumptions:

1. Gas molecules can physically adsorb in infinite successive layers.
2. Langmuir's monolayer theory applies to each individual layer.
3. The heat of adsorption for the first layer is E_1 (interaction with solid adsorbent).
4. For all subsequent layers (2nd, 3rd, . . . , nth), the heat of adsorption is constant and equal to the heat of liquefaction/condensation of the bulk gas (E_L).
5. Evaporation and condensation occur only between adjacent layers.

The BET Isotherm Equation:

$$\frac{P}{V(P_0 - P)} = \frac{1}{V_m C} + \frac{C - 1}{V_m C} \left(\frac{P}{P_0} \right)$$

- P : Equilibrium gas pressure
- P_0 : Saturation vapor pressure of the gas at measurement temperature
- V : Total volume of gas adsorbed at STP
- V_m : Volume of gas required to form a complete unimolecular monolayer at STP
- C : Dimensionless BET parameter related exponentially to net adsorption enthalpy:

$$C \approx \exp \left(\frac{E_1 - E_L}{RT} \right)$$

Surface Area Determination:

1. Plot $\frac{P}{V(P_0 - P)}$ on the y-axis against relative pressure $\frac{P}{P_0}$ on the x-axis (typically valid across linear range $0.05 \leq \frac{P}{P_0} \leq 0.35$).
2. Obtain slope $s = \frac{C-1}{V_m C}$ and y-intercept $i = \frac{1}{V_m C}$.
3. Solve simultaneously for monolayer volume V_m :

$$V_m = \frac{1}{s + i}$$

$$C = \frac{s}{i} + 1$$

4. Calculate the specific surface area (S_{BET}):

$$S_{BET} = \frac{V_m \cdot N_A \cdot \sigma}{V_{mol}}$$

Where N_A is Avogadro's number (6.022×10^{23} molecules/mol), σ is the molecular cross-sectional area of the adsorbate (e.g., 0.162 nm^2 for N_2 at 77 K), and $V_{mol} = 22,414 \text{ cm}^3 \cdot \text{mol}^{-1}$.

5. Applications of Adsorption

- **Ultra-High Vacuum Generation:** Activated charcoal cooled in liquid nitrogen (Dewar flasks) adsorbs residual traces of air to produce pressures down to 10^{-10} Torr.
- **Gas Masks and Personal Protective Equipment:** Activated carbon treated with metal oxides adsorbs toxic vapors (CO , Cl_2 , SO_2 , phosgene) while letting breathable air pass.
- **Industrial Humidity & Moisture Control:** Desiccants like silica gel and activated alumina selectively adsorb atmospheric moisture in packaging, electrical transformers, and compressed gas pipelines.
- **Decolorization & Purification in Industry:** Bone char and granular activated carbon remove colored impurities from raw cane sugar syrups, edible oils, and pharmaceuticals.
- **Heterogeneous Catalysis:** Solid catalysts provide active surfaces for reactant concentration and activation via chemisorption (e.g., finely divided Fe in the Haber ammonia synthesis; Pt/Rh in automotive catalytic converters).
- **Chromatographic Separation:** Analytical techniques (TLC, column chromatography, HPLC) exploit differential adsorption affinities to separate complex chemical and biological mixtures.

- **Froth Flotation Process:** Low-grade sulfide ores (PbS , ZnS) are selectively wetted by pine oil and collector molecules, adsorbing onto rising air bubbles to float away from gangue (SiO_2).
 - **Water Treatment and Ion Exchange:** Synthetic zeolite beds and polymeric resins demineralize water by selectively exchanging hard multivalent cations (Ca^{2+} , Mg^{2+}) via localized surface exchange.
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